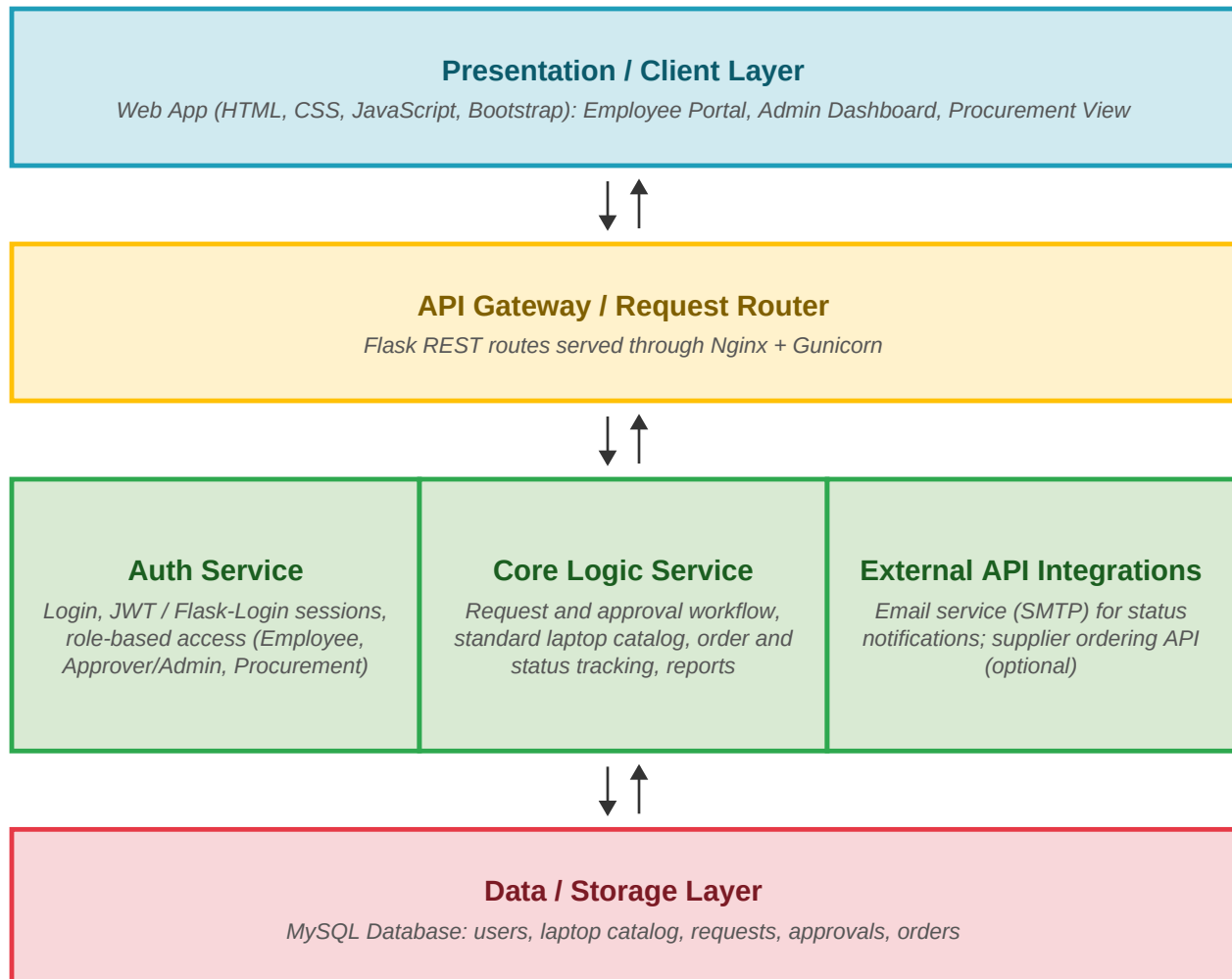


Solution Architecture

Date	15 March 2026
Team ID	PNT2022TMID124356
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level layout of the laptop procurement system: Presentation Layer, API Gateway, Application Services and Data Layer.



Request flow: Employee submits a laptop request from the standard catalog. Admin reviews and approves or rejects it. Approved requests become standard laptop orders placed by procurement. Status is tracked on the dashboard and requesters and admins are notified by email.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Browser-based interface where employees submit laptop requests and check status, approvers review and approve or reject them, and procurement staff view orders and the dashboard.	HTML5, CSS3, JavaScript, Bootstrap
API Gateway	Single entry point that receives every client request, routes it to the right service and returns the response.	Flask (REST routes), Nginx, Gunicorn
Auth Service	Handles login, session/token validation and role-based access so each user sees only what their role allows.	Flask-Login, JWT, Werkzeug password hashing
Core Logic Service	Runs request submission, approval/rejection, the standard laptop catalog, order creation, status tracking and basic reports.	Python, Flask, SQLAlchemy
External API Integrations	Sends status-update notifications to requesters and admins; optionally passes standard orders to a supplier.	SMTP / Flask-Mail, supplier API (optional)
Database	Stores users, the standard laptop catalog, requests, approval history, orders and their status.	MySQL